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Testing of 3D printed vs. fiberglass trapezoidal flumes
Submitted to: CSU Agricultural Water Quality Program

1. Objective

The purpose of this report is to summarize the results of trapezoidal flume testing conducted at the CSU Hydraulics Laboratory for the CSU Agricultural Water Quality Program. This testing aimed to evaluate the hydraulic performance and flow rating accuracy of 3D-printed trapezoidal flumes under controlled conditions.

2. Methods

Volumetric flow rate and flow depth measurements were made at the CSU hydraulics lab to test the performance of a 3D printed trapezoidal flume used in agricultural applications. The equipment used to measure the flow rate and flow depth have accuracy reported in Table 1.

Table 1. Equipment accuracy

Equipment	Accuracy
Magnetic flow meter	+/- 0.02%
Point gauge	+/- 0.005 (ft)
Flume infiltration	+/- 0.02 cfs

Photos of the flume during testing are shown in Figure 1. Many additional photos and videos were collected during testing which were shared with the Agricultural Water Quality Program point of contact, Emmanuel Deleon via a OneDrive folder. Depth and flow measurements were also shared with the CSU AWQP in an Excel spreadsheet titled “3d_printed_flume_data.xlsx”.



Figure 1. Measurements of flow depth in 3D printed v-notch flume.

3. Results

The measured depths and flow rates for each of three flumes is reported in Tables 2-4. The reported point gauge reading in each of the tables is the value measured in the Hydraulics Laboratory. The depth (ft) column is the point gauge reading minus the flume bottom point gauge reading.

Table 2. Measured data in 3D Printed Flume #1

	3D Printed Flume #1		
	Point Gauge Reading	Depth (ft)	Flow rate (cfs)
	2.737	0.371	0.1
	2.695	0.413	0.15
	2.651	0.457	0.2
	2.612	0.496	0.25
	2.58	0.528	0.3
	2.565	0.543	0.32
Bottom	3.108		

Table 3. Measured data for Fiberglass Flume

	Fiberglass Flume		
	Point Gauge Reading	Depth (ft)	Flow rate (cfs)
	2.047	0.326	0.04
	2	0.373	0.08
	1.96	0.413	0.12
	1.842	0.531	0.26
Bottom	2.373		

Table 4. Measured data for 3D Printed Flume #2

	3D Printed Flume #2		
	Point Gauge Reading	Depth (ft)	Flow rate (cfs)
	2.861	0.241	0.03
	2.774	0.328	0.08
	2.711	0.391	0.13
	2.668	0.434	0.18
	2.621	0.481	0.23
	2.581	0.521	0.28
Bottom	3.102		

A power function was fit to the flow vs stage data. The power trendlines fit these data well ($R^2 \cong 1$), as expected for trapezoidal flumes (See Figure 2). There is a greater difference in predicted discharge at low depths (stage < 0.4) when comparing the fiberglass and 3D printed flumes because of the slight difference in hydraulic roughness of the construction material. However, with increasing depth, the reported flow rate for both the 3D printed and fiberglass trapezoidal flumes appears to converge.

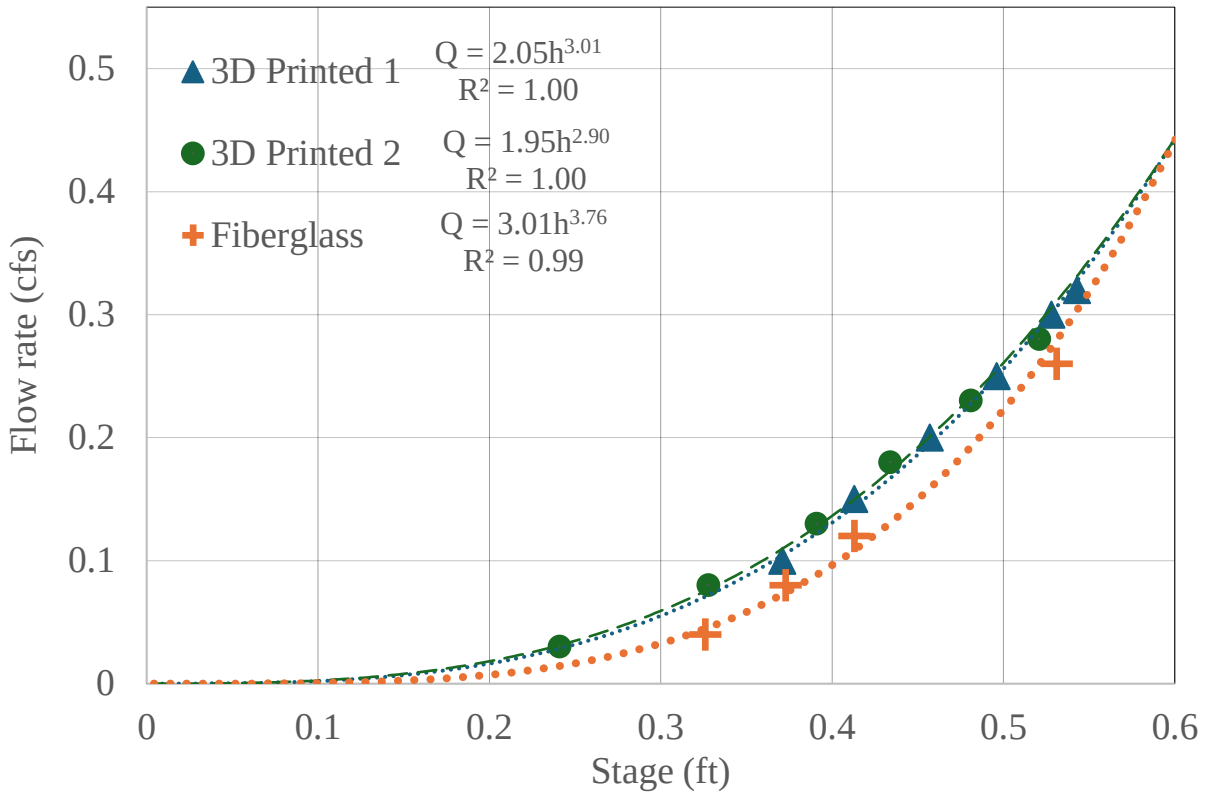


Figure 2. Stage-discharge relationship and fitted power trendlines for both the 3D printed trapezoidal flume and the fiberglass trapezoidal flume. Both flumes performed similarly.

4. Conclusions and recommendations

During testing, one of the 3D printed trapezoidal flumes was damaged and may be an indication of the materials susceptibility to wear and tear during installation in agricultural or other applications. Hydraulically, however, the 3D printed flume performed with reasonable accuracy. The AWQP has suggested they may be interested in testing the 3D printed flumes in a field setting over longer time scales, which the CSU Hydraulics Lab is willing and available to conduct. We recommend comparing the fitted curves with those reported by the fiberglass trapezoidal flume manufacturer to assess the relative accuracy of flow measurement for all three flumes tested by the Hydraulics Laboratory. The fiberglass flume manufacturer should provide a power function such as those reported in Figure 2 such that flow rate, Q is reported as:

$$Q = ah^b$$

where a and b are empirical coefficients, and h is measured depth. We recommend that a statistical analysis be conducted to compare measured coefficients in these tests with those reported by the manufacturer. The CSU Hydraulics Laboratory is happy to support the AWQP with this and any future flume testing, calibration, or related hydraulic assessments.